

Malte Christoph Rühlemann

Postdoctoral Fellow

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University training and degree

- 2015 - 2020 Dr. rer. nat. Biology, Kiel University, Germany. Supervisor: Prof. Dr. Andre Franke, Title: 'The interaction of genetics, inflammation and the microbiome in the human metaorganism', Grade: *summa cum laude*
- 2012 - 2014 M. Sc. Medical Life Sciences, Kiel University, Germany
- 2008 - 2011 B. Sc. Life Science, University of Hannover, Germany

Academic and Research Experience

- 2024 - Project PI within the DFG-funded CRC1182 'Origin and Function of Metaorganism', Project C5.3 'Functional consequence of the Prevotella-to-Bacteroides enterotype shift in the human gut microbiome.'
- 2024 - 2025 *Parental leave: 7 months (Oct 2024 + Apr-Oct 2025)*
- 2023 - Postdoctoral Researcher, Genomics and functions of host-microbiome systems w/ Prof. Mathieu Groussin & Prof. Mathilde Poyet (Groussin & Poyet Lab), Institute of Clinical Molecular Biology (IKMB), UKSH & CAU Kiel, Kiel, Germany.
- 2021 - Guest Postdoctoral Researcher at the Institute for Medical Microbiology and Hospital Epidemiology w/ Prof. Dirk Schlüter & PD Marius Vital, Hannover Medical School, Hannover, Germany.
- 2020 - 2022 Postdoctoral Researcher, Genetics & Bioinformatics w/ Prof. Andre Franke, Institute of Clinical Molecular Biology (IKMB), UKSH & CAU Kiel, Kiel, Germany.
- 2020 - 2021 Guest Postdoctoral Researcher at Bioinformatics and (Eco)-Systems Biology Lab w/ Prof. Jeroen Raes, VIB/KU Leuven, Leuven, Belgium.

Involvement in large collaborative research projects

- 2024 - EU-funded miGut-Health project. Role: I am leading the data processing and meta-analysis of a dataset combining metagenomic sequencing data of > 5,000 IBD-affected individuals and >5,000 non-IBD controls as the central analysis of WP1.
- 2024 - DFG CRC 1182 'Origin and Function of Metaorganisms', 3rd funding phase: PI in Project C5 'Functional impact of recent shifts in human lifestyle on host-microbiome interactions.' Role: Project design and writing of application (C5.3) and co-design and co-writing of project (A2.2).
- 2023 - The Global Microbiome Conservancy (GMbC), a project aimed at the conservation of global human-associated microbiomes, while at the same time aiming for the involvement of and capacity building in underrepresented human communities. Role: Analysis lead in the flagship scientific articles centered on host-genomic, environmental, and lifestyle influences on the gut microbiome.
- 2021 STORMS Consortium for the development of standardized reporting formats in microbiome research. Role: Contributor.
- 2020 - 2023 DFG CRC 1182 'Origin and Function of Metaorganisms', 2nd funding phase: Postdoctoral researcher in Project A2 'Host-microbiome coevolution in the mammalian intestine', PI

	Franke. Role: Co-design of project, co-writing of application and conduction of research project, elected junior representative on the steering committee.
2020 - 2023	The Severe COVID-19 Study Group and COVID-19 Host Genetics Initiative. Role: Contributor.
2019 -	DFG RU 5042 'miTarget: the microbiome as a therapeutic target in inflammatory bowel disease'. Role: Project support for INF 'Data management, bioinformatics and statistics' and P2 'Ecology and function of synthetic bacterial communities to understand and modulate IBD-associated microbiomes'. Since 2024: Co-supervisor of PhD Maryna Gozhenko
2018 - 2021	MiBioGen consortium for microbiome GWAS / QTL analysis. Role: Data analysis lead and main contact for Kiel-based cohorts (FOCUS and BSPSPC).
2016 - 2019	DFG CRC 1182 'Origin and Function of Metaorganisms', 1 st funding phase: Doctoral candidate in Project A2 'Host-microbiome coevolution in the mammalian intestine', PI Franke. Role: conduction of research project.
2016 - 2019	DFG CRU 306 'Primary Sclerosing Cholangitis (PSC)', UKE Haubug, 1 st funding phase: Data analyst in Project P1 'The intestinal and biliary microbiome in PSC', PI Franke & Schramm. Role: Data analysis lead.
2014 - 2015	BMBF e:Med 'sysINFLAME': Doctoral candidate in Project CP3 'Host-genetics meets microbiome', PI Franke.

Third party funding and awards

2024 - 2027	DFG Collaborative Research Center 1182 'Origin and Function of Metaorganisms', Third Funding Phase (total funding volume: 11,838,101); PI in subproject C5.3: 'Functional consequence of the <i>Prevotella</i> -to- <i>Bacteroides</i> enterotype shift in the human gut microbiome'; Subproject funding volume: 277,465€. Note: I was chosen as one of two postdoctoral researchers from the second funding phase to develop an independent research project as principal investigator in a multi-step selection process.
2026	Shortlisted for the zukunft.niedersachsen "AI Research Groups Lower Saxony" program (VolkswagenStiftung & Lower Saxony Ministry of Science and Culture); invited for presentation to the panel for May/June 2026.
2024	Invited participant for the first HUMAN-MICROSTAR selection symposium @ HZI Braunschweig.
2023	Bruhn scholarship for the promotion of early career researcher (Förderpreis der Bruhn-Stiftung): 40,000€
2021 - 2024	DFG Sequencing Call (Sachmittelbeihilfe) 'Identification of microbiome-host genetics associations using full-length 16S rRNA gene sequencing'; Funding volume: 364,675€
2020	Kiel Life Science Postdoc Award: 6,000€
2017	CRC1182 Young Investigator Award: 15,000€
2017	ZMB Young Scientist Grant for Doctoral Students: 10,000€

Teaching experience

I have supported the supervision of multiple Master's and PhD students in my role as PhD student and PostDoc at the IKMB, especially concerning statistical data analysis focused on – but not limited to – microbiome data. I currently supervise PhD student Hina Ayub (CRC1182), co-supervise PhD student Maryna Gozhenko (miTarget), and substantially support PhD students Eike Matthias Wacker (miTarget), Johanna Saalfrank (CRC1182), and Zaira Zafroon (miTarget).

- 2022 Workshop Instructor for 'Basic Microbiome Analysis in R' at the Joint Summer School 'Frontiers in Microbiome Research' of the Danish Diabetes Academy, University of Copenhagen and Kiel University Ebberup, Denmark.
- 2021 Organizer and Instructor 'Basic R and intro to 16S rRNA gene analysis workshop' for PhD candidates and PostDocs in the CRC1182 'Origin and Function of Metaorganisms', Kiel.
- 2019 Workshop Instructor for 'Best Practice in Bioinformatics of Microbiome Data' at the Autumn School 'The Role of the Microbiome in Complex Metabolic & Inflammatory Diseases' of the Danish Diabetes Academy, University of Copenhagen and Kiel University, Ebberup, Denmark.
- 2018 Organizer and Instructor '16S Amplicon Sequencing Workshop' for PhD candidates and PostDocs in the CRC1182 'Origin and Function of Metaorganisms', Kiel.
- 2016 - Master's Programme Medical Life Sciences, Kiel University, Bioinformatics lecture (support) and hands-on workshop: 'Introduction to Microbiome Analysis'.

Invited talks

- 2025 'Stay Healthy Cluster Symposium' of the EU-funded IMMEDIATE project, representing the miGut-Health project: 'Microbiome in IBD: new insights from the international miGut-Health meta-analysis'
- 2025 Annual conference of the 'Deutsche Gesellschaft für Mukosale Immunologie und Mikrobiom': 'Mikrobiom in CED: Neue Erkenntnisse aus der KINDRED Familienkohorte und der miGut-Health Meta-Studie'
- 2022 Workshop 'Biobanking in Microbiome Research' of the German Biobank Node: 'GWAS for the gut microbiome'
- 2021 SymbNET Kick-off Meeting 'Host-microbiota interactions: Insights from human genetics and primate evolution'
- 2021 NGS Competence Network (DFG funded): 'Host-microbiota interactions: Insights from human genetics and primate evolution'
- 2021 Berchtesgaden Microbiome Science Days: 'Probiotics from Primates? How human genetics and evolution shape the human metaorganism'
- 2018 Workshop of the DFG Research Group 'Sociality and Health in Primates': 'Non-invasive markers for inflammation'
- 2018 Berchtesgaden Microbiome Science Days: 'Investigating evolutionary and environmental factors shaping the intestinal microbiota'
- 2018 CRC1182 Young Investigators Research Day: 'Evolutionary and environmental factors shaping the gut microbiome in humans and non-human primates'
- 2017 Life Science Student Conference: 'The effect of captivity on the community composition and functional potential of the intestinal microbiota of chimpanzees (*Pan troglodytes*)'
- 2016 Berchtesgaden Microbiome Science Days: 'Gut microbial communities as potential biomarkers in Primary Sclerosing Cholangitis'

Regular progress reports within the CRC1182 PhD candidate seminars. Talks at annual retreats of the CRC1182 (2016-2025) and KFO306 (2016-2019).

Professional development

- 2022 - 2023 Professional Leadership in Science: three 2-day courses of the modules 'Lead Self', 'Lead Team', and 'Lead Organisation', Center for Science & Research Management (ZWM Speyer)
- 2020 CRC1182 workshop participant "Networking competence"

2017 - 2018 Professional Coaching of CRC1182 high-potential doctoral students by Ute Jülly (www.juellys.com)

Outreach

2025 DFG forschung magazine, 1/2025: Invited article on the connection between inflammatory bowel disease and vanishing microbes in the human gut: 'Verschwundene Mikroorganismen – Evolutionäre Medizin: Entschlüsselung chronisch entzündlicher Darmerkrankungen'.

2024 Interviewed for a press release of Kiel University following the publication of my 2024 article on the great ape microbiome in *Nature Communications* (<https://www.uni-kiel.de/de/detailansicht/news/003-ruehlemann-natcomms>)

2023 Invited interview 'Discussing the Future of Kiel Life Science' and panelist at round table discussion.

2022 Panelist at CRC1182 Round Table Discussion 'Future of Metaorganism Research'

2022 Talk at the 'Darwin Day 2022: Revolutions in Biology': 'The evolution of the gut microbiom' for over 1,000 school children (+ online participants) from Schleswig-Holstein in the larges lecture hall of CAU Kiel.

2021 Pint of Science Kiel: 'Probiotics from Primates? An evolutionary perspective on the human metaorganism'

2021 Radio Interview for NDR Info following the publication of 'Genome-wide association study in 8,956 German individuals identifies influence of ABO histo-blood groups on gut microbiome' in *Nature Genetics*

Leadership and service

IKMB Organization of the group's regular meetings including progress reports and journal clubs, co-mentoring and support of PhD candidates and Master's students, supervision of visiting scientist Dr. Serena Piticchio (University of Barcelona)

CRC1182 Project PI (C5.3) since 2024, supervising PhD Student Hina Ayub. Elected junior representative and member of the Steering Committee (2nd funding phase: 2020 - 2023), co-organizer and session chair at the 'Young Investigators Research Day 2018'.

Peer review Genome biology and evolution, International Journal of Obesity, Scientific Reports, Nature Genetics, Microbiome, Nature Ecology & Evolution, Frontiers in Immunology